

DAY 3

Part 1: Interview Warm-Up

1. What is a Validation Rule?

A Validation Rule is a Salesforce feature that ensures data entered into a record follows predefined business rules before the record is saved. If the rule evaluates to TRUE, Salesforce blocks the save operation and displays an error message.

Example:

Prevent a student from submitting a job application if their CGPA is lower than the minimum CGPA required

2. What is a Flow?

A Flow is Salesforce's declarative automation tool that allows you to automate business processes without writing Apex code. Flows can create, update, or delete records, send emails, and perform various actions based on specific conditions.

Example:

Automatically populate the Application Date when a new application is submitted.

3. What is an Apex Trigger?

An Apex Trigger is a piece of Apex code that runs automatically before or after database events such as insert, update, delete, or undelete. Triggers are used when business requirements are too complex to implement using declarative tools like Flows.

Example:

Prevent duplicate job applications or update related records after an interview is completed.

4. When would you choose a Flow instead of an Apex Trigger?

A Flow is the preferred choice when the automation is straightforward and does not require complex programming logic. Since Flows are declarative, they are easier to build, maintain, and modify.

Common use cases:

Automatically populate fields

Send email notifications

Create or update related records

5. Can a Validation Rule update another field? Why?

No. A Validation Rule is designed only to validate data before a record is saved. It cannot modify field values. Its only purpose is to either allow the save operation or stop it by displaying an error message.

To automatically update fields, use a Flow or an Apex Trigger

6. Which executes first: Validation Rule, Flow, or Trigger?

For record creation or update, the simplified execution order is:

Before-Save Record-Triggered Flow

Before Trigger

Validation Rule

Record is saved

After Trigger

After-Save Record-Triggered Flow

7. What is a Record-Triggered Flow?

A Record-Triggered Flow is a Flow that automatically runs whenever a record is created, updated, or deleted. It helps automate business processes without writing Apex code.

Example:

When a new Application record is created, automatically set the Application Date and send an email notification to the Placement Officer.

Part 2: Business Scenario

Scenario

You are developing a Placement Management System. Choose the appropriate Salesforce automation tool for each requirement and explain why.

1. Send an email to the Placement Officer when a student submits an application.

Solution: Flow

Reason:

An After-Save Record-Triggered Flow can automatically send an email whenever a new Application record is created. This is a standard automation task that does not require Apex code.

2. Automatically populate the Application Date.

Solution: Flow

Reason:

A Before-Save Record-Triggered Flow can populate the Application Date with the current date before the record is saved. This approach is efficient and performs better than Apex.

3. Prevent duplicate applications for the same student and job.

Solution: Apex Trigger

Reason:

An Apex Trigger can check existing Application records before inserting a new one. If a duplicate application exists for the same student and job, the trigger prevents the record from being saved.

4. Reject applications when the student's CGPA is below the job's minimum requirement.

Solution: Validation Rule

Reason:

A Validation Rule verifies the student's CGPA before the record is saved. If the CGPA does not meet the required criteria, Salesforce displays an error message and blocks the save operation.

5. Automatically create an Offer Letter when the Application Status changes to Selected.

Solution: Flow

Reason:

A Record-Triggered Flow can detect when the Status changes to Selected and automatically create a related Offer Letter record without writing Apex.

Summary Table

Requirement	Solution	Reason
Send email notification	Flow	Automates email notifications without code
Auto-fill Application Date	Flow	Populates the field before the record is saved
Prevent duplicate applications	Apex Trigger	Performs complex duplicate validation
Reject applications with low CGPA	Validation Rule	Validates business rules before saving
Create Offer Letter	Flow	Automatically creates related records when the status changes

Part 3: Design Challenge

Choose the most suitable Salesforce feature for each requirement.

Requirement	Solution	Reason
Automatically set the Application Date when a new application is created	Flow	A Before-Save Record-Triggered Flow populates the field before the record is saved.
Prevent students with insufficient CGPA from applying	Validation Rule	Ensures the CGPA meets the minimum requirement before saving the record.
Prevent duplicate applications for the same student and job	Apex Trigger	Checks existing records and prevents duplicates using <code>addError()</code> .
Send an email to the Placement Officer after an application is submitted	Flow	An After-Save Record-Triggered Flow automatically sends the notification.
Create an Offer Letter when the application status becomes Selected	Flow	Detects the status change and creates the related Offer Letter automatically.
Update multiple related records using complex business logic	Apex Trigger	Provides flexibility, supports bulk processing, and handles complex operations efficiently.

Part 4: Hands-on Task

Objective

Create a Record-Triggered Flow for the Placement Management System that:

Automatically populates the Application Date.

Sends an email notification to the Placement Officer.

Executes successfully whenever a new Application record is created.

Implementation

Step 1: Auto-Populate the Application Date

Flow Type: Before-Save Record-Triggered Flow

Object: Application

Trigger: When a record is created

Assignment:

Application_Date__c = \$Flow.CurrentDate

This ensures that every newly created Application record automatically stores the current date without requiring user input.

Step 2: Send an Email Notification

Flow Type: After-Save Record-Triggered Flow

Trigger: When an Application record is created.

Action: Send an email to the Placement Officer.

Email Content:

Student Name

Job Title

Application Date

Application Status

This keeps the Placement Officer informed whenever a new application is submitted.

Step 3: Verify the Flow

Create a new Application record.

Save the record.

Verify that:

The Application Date is populated automatically.

An email notification is sent to the Placement Officer.

The Flow completes successfully without errors.

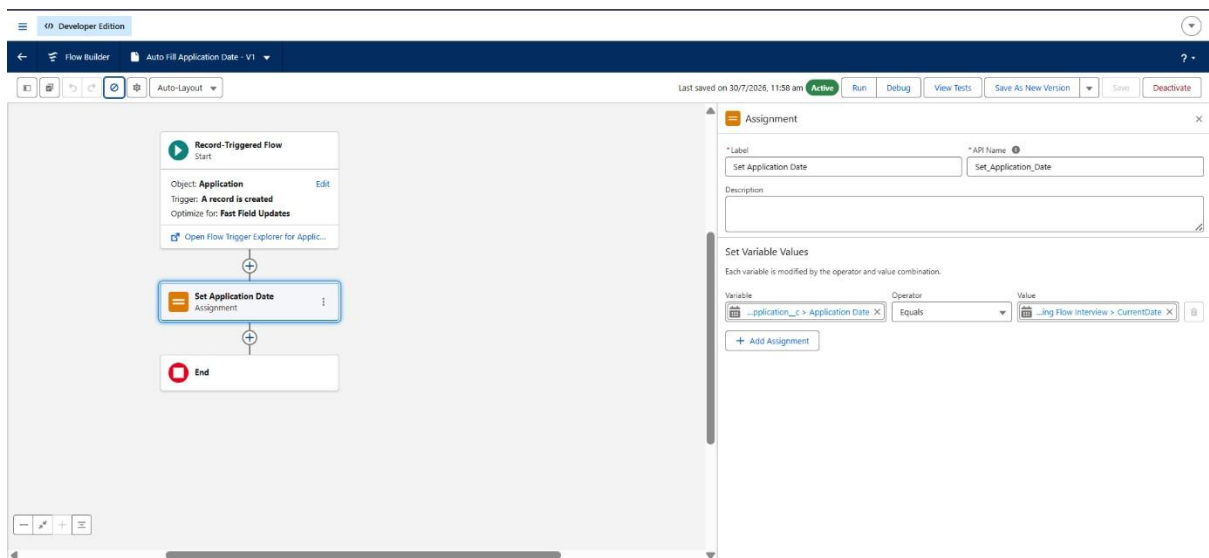
Key Takeaways

Validation Rules validate data and prevent invalid records from being saved.

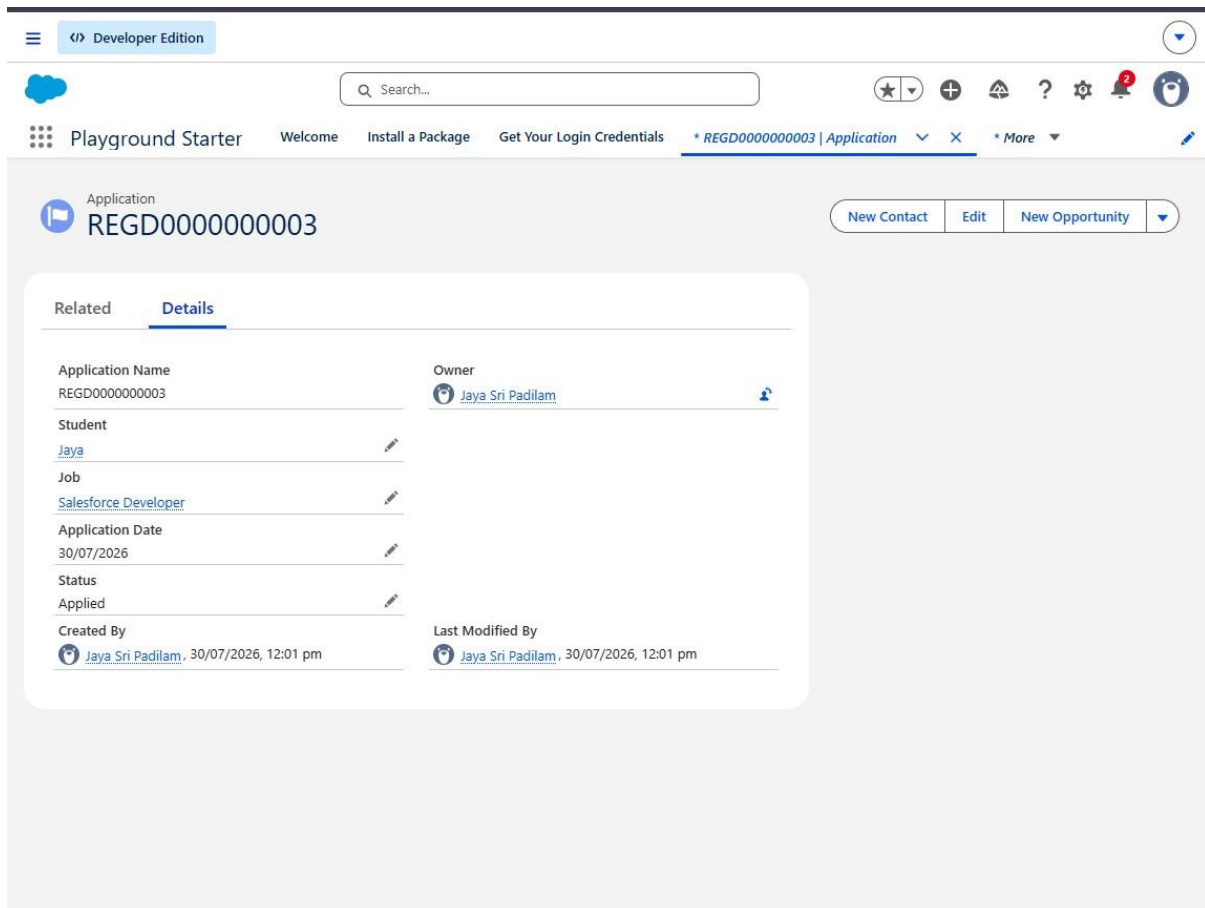
Flows automate routine business processes without coding.

Apex Triggers are used for complex business logic and advanced record processing.

Selecting the appropriate automation tool improves system performance, maintainability, and scalability.



Whenever a student submits a new application, the Application Date is automatically set to the current date.



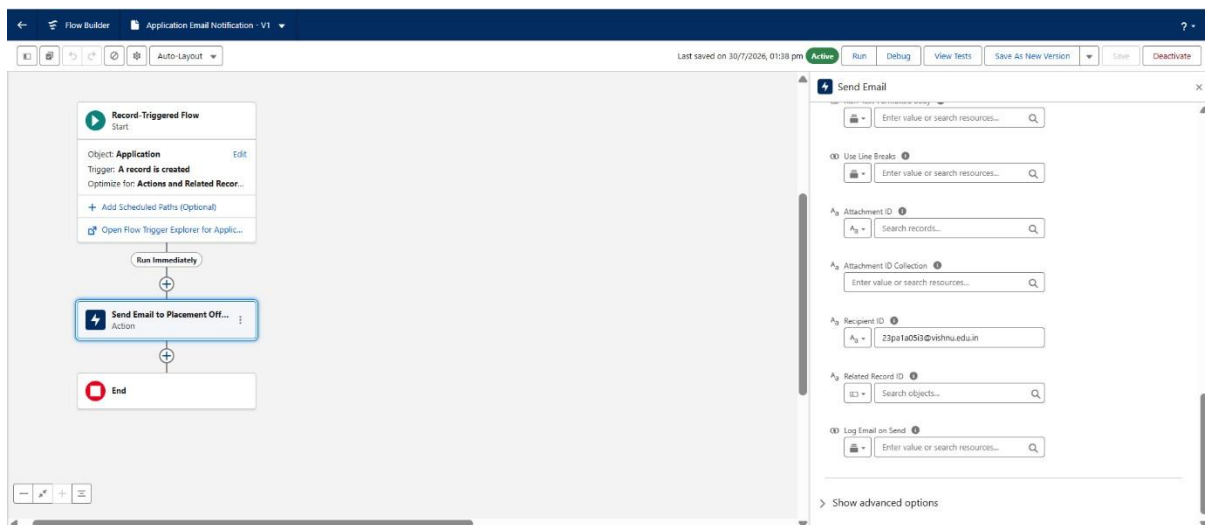
Email Notification

Type: After-Save Record-Triggered Flow

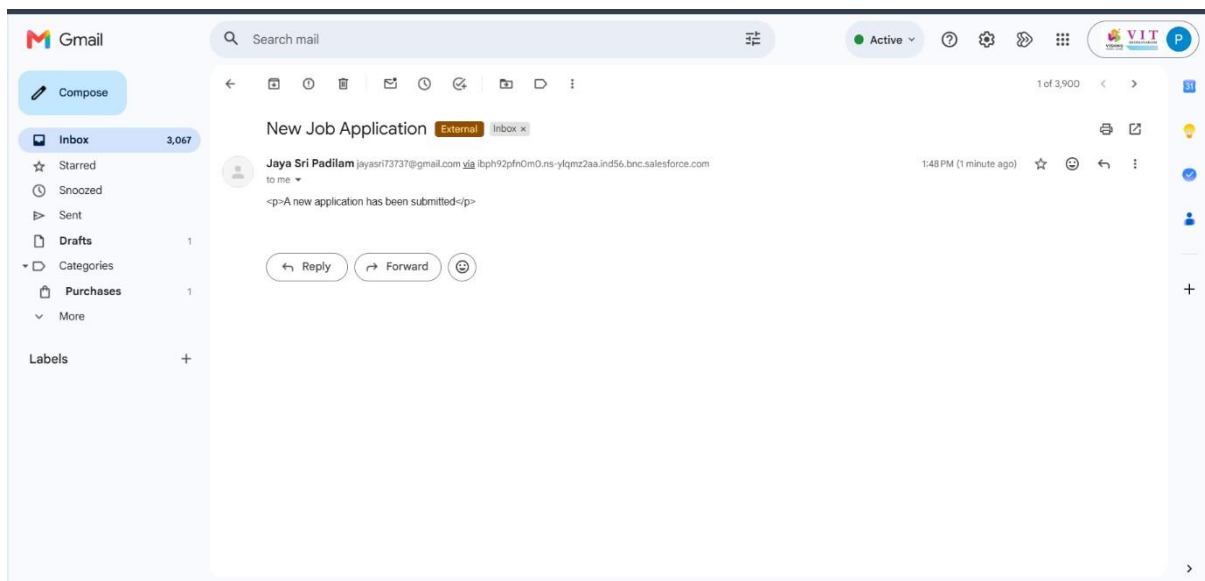
Object: Application

Trigger: When a record is created

Action: Send Email



The Placement Officer is automatically notified whenever a new application is submitted.



Part 5 – Hands-on Task: Validation Rules

Objective

Create Validation Rules in the Application object to ensure only valid applications are submitted. Validation Rules improve data quality by preventing invalid records from being saved

Validation Rule 1 – CGPA Check

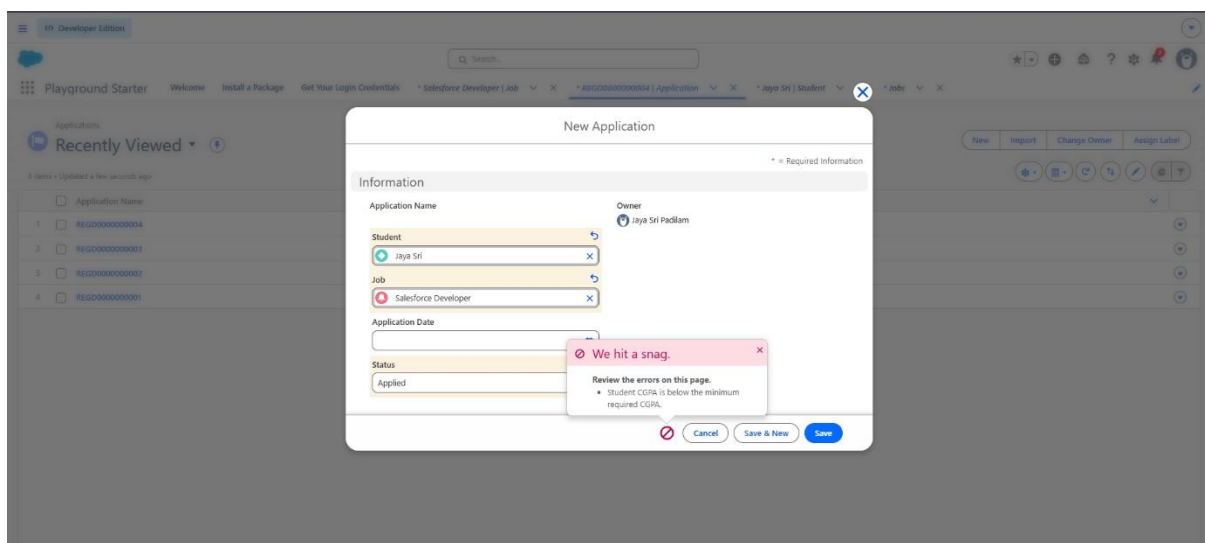
Rule Name: CGPA_Check

Purpose:

Prevent students from applying if their CGPA is less than the minimum CGPA required for the selected job.

Formula:

$\text{Student_r.CGPA_c} < \text{Job_r.Minimum_CGPA_c}$



Validation Rule 2 – Closing Date Check

Rule Name: Closing_Date_Check

Purpose:

Prevent students from applying after the job's closing date.

Formula: Application_Date__c > Job__r.Closing_Date__c

The screenshot shows the Salesforce Setup interface. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'Application' and shows a list of 'Validation Rules'. There are three rules listed: 'CGPA_Check', 'Closing_Date_Check', and 'Mandatory_Fields'. Each rule has columns for Rule Name, Error Location, Error Message, Active status, and Modified By. The 'Closing_Date_Check' rule is highlighted.

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
CGPA_Check	Top of Page	Student CGPA is below the minimum required CGPA.	✓	Jaya Sri Padilam, 30/07/2026, 1:52 pm
Closing_Date_Check	Top of Page	Application date cannot be after the job closing date.	✓	Jaya Sri Padilam, 30/07/2026, 1:55 pm
Mandatory_Fields	Top of Page	Student and Job fields are mandatory.	✓	Jaya Sri Padilam, 30/07/2026, 1:56 pm

Validation Rule 3 – Mandatory Fields

Rule Name: Mandatory_Fields

Purpose:

Ensure that both the Student and Job fields are selected before saving an application.

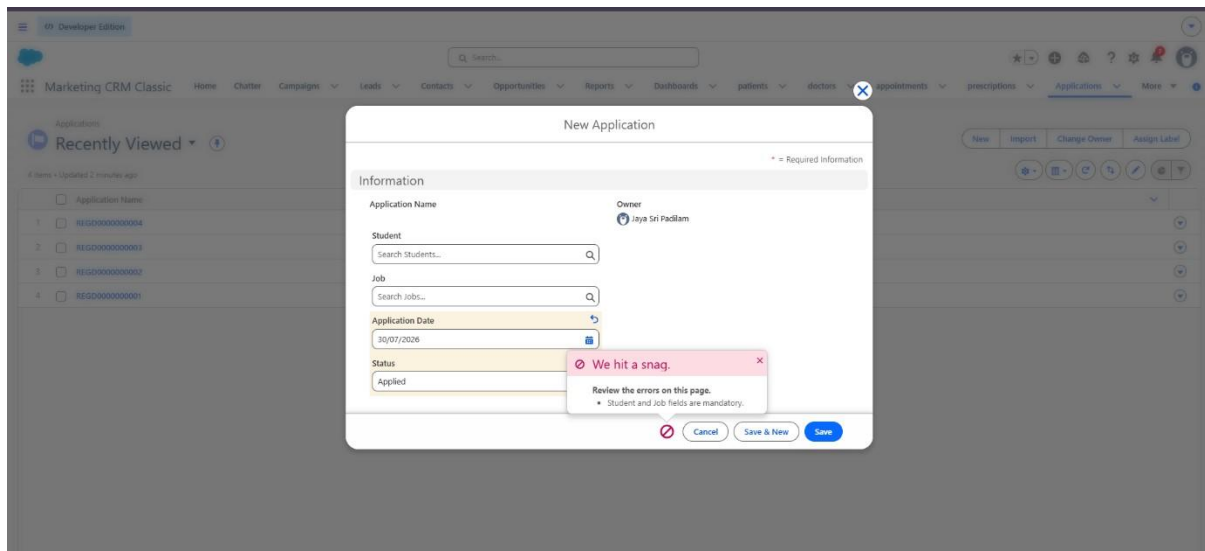
Formula:

OR(

ISBLANK(Student__c),

ISBLANK(Job__c)

)



Part 7 – Mini Project Enhancement

Objective

Enhance the Placement Management System by implementing Record-Triggered Flows and Validation Rules to automate the application process and improve data accuracy.

1. Auto-Populate Application Date

Implementation

Created a Before-Save Record-Triggered Flow on the Application object.

Configured the flow to automatically populate the Application Date with the current date whenever a new application is created.

Result

The Application Date is automatically filled, eliminating manual data entry and ensuring accurate application records

2. Email Notification to the Placement Officer

Implementation

Created an After-Save Record-Triggered Flow on the Application object.

Configured the flow to send an email notification to the Placement Officer whenever a student submits a new application.

Result

The Placement Officer receives an automatic email for every new application, enabling timely review and follow-up.

3. Validation Rules

Implementation

The following validation rules were implemented to improve data quality:

CGPA Validation: Prevents students from applying if their CGPA is below the minimum CGPA required for the selected job.

Closing Date Validation: Blocks applications submitted after the job's closing date.

Mandatory Fields Validation: Ensures that the Student and Job fields are completed before the record can be saved.

Result

Only valid and complete application records are stored in Salesforce, reducing data errors and maintaining consistency.

4. Automatic Offer Letter Creation

Implementation

Created a Record-Triggered Flow on the Application object.

Configured the flow to execute whenever the Application Status changes to Selected.

Added a Create Records element to automatically create a related Offer Letter record.

Result

The flow was successfully created and activated. However, during testing, the Offer Letter record was not generated, indicating that further troubleshooting and debugging are required.

Part 8 – Debugging Challenge

During testing of the Placement Management System, some automation processes did not behave as expected. The objective was to identify the issue, determine its possible causes, and apply appropriate debugging techniques to resolve the problem.

Challenge 1 – Offer Letter Flow Not Creating Records

Problem

The Offer Letter record was not created automatically when the Application Status was updated to Selected.

Possible Causes

The flow entry conditions were configured incorrectly.

Required fields on the Offer Letter object were not populated.

Incorrect field mapping was used in the Create Records element.

The flow did not execute because the record failed to meet the specified entry criteria.

Debugging Approach

To identify the root cause, the following checks were performed:

Verified that the flow was activated.

Reviewed the entry conditions to ensure the flow runs only when the status changes to Selected.

Checked the field mappings in the Create Records element.

Confirmed that all required Offer Letter fields were assigned valid values.

Used Flow Debug and Debug Logs to trace the flow execution and identify any errors.

Outcome

The investigation helped narrow down the potential causes of the issue. Additional debugging and configuration updates are required before the Offer Letter can be created successfully.

Key Learnings

Before-Save Flows are ideal for updating fields efficiently before records are saved.

After-Save Flows are commonly used for sending notifications and creating related records.

Validation Rules help maintain data integrity by preventing invalid records from being saved.

Flow Debugging and Debug Logs are valuable tools for diagnosing automation issues and identifying configuration errors.